

P14 n balls tossed ~~independently~~ (indp. and random)
into n boxes.

Aim: Find prob. that exactly one box is empty.

One box is empty $\Leftrightarrow$ Exactly one box contains 2 balls, ~~and~~
the remaining $n-2$ boxes contain 1 ball,
and one box is empty.

$\therefore$ The number of such outcomes

$$= n \cdot (n-1) \cdot \binom{n}{2} \cdot (n-2)!$$

Here n is the number of ways to choose a box to be empty,

* $\binom{n-1}{1}$ is the # of ways to choose one of the remaining
 $n-1$ boxes to contain 2 balls.

* $\binom{n}{2}$ is the # ways to select two balls that are tossed into a box.

* $(n-2)!$ is the # ways to ~~to~~ arrange $n-2$ remaining balls into $n-2$ remaining boxes.

Since there are n^n ways to ~~to~~ toss n balls into n boxes (repetition is allowed),

the prob. is
$$\frac{n \binom{n-1}{1} \binom{n}{2} (n-2)!}{n^n} = \frac{n! \binom{n}{2}}{n^n}.$$